SEUNGWOO LYU

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EDUCATION

Korea University, Seoul, South Korea Mar 2021 - Feb 2027 (Expected)
B.S. in Smart Security (Double Major in Data Science)
GPA: 4.32 / 4.5 (98.2 / 100)

University of Wisconsin-Madison, Madison, WI, USA Spring 2026
Exchange Student, Computer Sciences
GPA: 4.0 / 4.0

RESEARCH EXPERIENCE

NAVER, Physical Intelligence Agent Dev Team Jul 2026 - Aug 2026
Software Developer Intern
• Built end-to-end RAG and dialogue evaluation pipelines for a robot operations AI agent.

Multimodal Interactive Intelligence Laboratory, Korea University Dec 2024 - Present
Undergraduate Research Intern
• Conducted research on multimodal reasoning, graph-based data synthesis, and diffusion language models.
• Built automated evaluation and quality-check pipelines for safety and reasoning datasets.
• Worked on preference optimization and scalable training pipelines for large language models.
• Co-authored a CVPR 2026 paper on graph-based multimodal data synthesis for cross-modal reasoning.

PAPERS

CRIT: Graph-Based Automatic Data Synthesis to Enhance Cross-Modal Multi-Hop Reasoning
Second Author · CVPR 2026 · June 2026

Elastic-MDM: Efficient Masked Diffusion Models with Variable Sequence Lengths
First Author

Direct Diffusion Score Preference Optimization via Stepwise Contrastive Policy-Pair Supervision
Second Author · ECCV 2026 · September 2026

Revisiting Task Vectors: A Structural Analysis of Their Limits
Co-Author · Accepted to Findings of EMNLP 2026 · October 2026

OpenSafeIntent: Evaluating Intent-Calibrated Safe Completion Across Dual-Use Prompt Sets
Second Author · ARR August 2026 Review Completed · Not Yet Committed

TECHNICAL SKILLS

Languages: Python · C · C++ · Go · SQL
Frameworks: PyTorch · TensorFlow · Hugging Face Transformers · vLLM
Tools: Git · Linux · Docker · HTCondor
Areas: Multimodal Learning · Diffusion LLM · Diffusion Models · LLM · In-Context Learning

HONORS & SCHOLARSHIPS

• Academic Excellence Scholarship - Spring 2025
• Academic Merit Scholarship - Fall 2025
• Korea-U.S. Youth Exchange Program in Advanced Fields (4th Cohort) - 2026
• Kaggle - Home Credit: Credit Risk Model Stability - Silver Medal (122 / 3,856)
• Kaggle - LLM Prompt Recovery - 260 / 2,175

EXTRACURRICULAR ACTIVITIES

LG AImers (5th Cohort), LG Jul 2024 - Sep 2024
Google Machine Learning Bootcamp, Google Sep 2023 - Nov 2023
42 Seoul Innovation Academy Sep 2021 - Aug 2022
Cykor (Cybersecurity Club) Mar 2021 - Sep 2022

MILITARY SERVICE

Republic of Korea Air Force - Cyber Operations Center Sep 2022 - Jun 2024
Sergeant (Honorable Discharge)